

Matrix diagonalization

Key Points

To diagonalize a matrix, M first find its eigenvalues and eigenvectors

Then form a diagonal matrix with its eigenvalues

$$D = \text{diag}(\lambda_1, \lambda_2, \lambda_3)$$

Then form a matrix from column eigenvectors

$$U = \begin{pmatrix} \vdots & \vdots & \vdots \\ v_1 & v_2 & v_3 \\ \vdots & \vdots & \vdots \end{pmatrix}$$

$$\text{Then } M = UDU^{-1}$$

Basic Skills

Q1. Find the inverse of the following matrix $M = \begin{pmatrix} 5 & 3 \\ -4 & 2 \end{pmatrix}$

Q2. Explain how to work out the adjugate of a 3×3 matrix

Q3. Find the eigenvalues and eigenvectors for $M = \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix}$

Q4. Given that a 2×2 matrix has eigenvalues $\lambda = 1, 2$ with corresponding eigenvectors $v = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ write the matrix in diagonal form

Q5. Using the identity matrix explain what repeated eigenvalues mean when finding eigenvectors

Competency Skills

Q6. Diagonalize the following matrices

a) $\begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix}$ b) $\begin{pmatrix} 7 & -3 \\ -2 & 6 \end{pmatrix}$

Q7. By diagonalizing the matrix $M = \begin{pmatrix} 2 & 6 \\ 0 & -1 \end{pmatrix}$ work out M^4

Q8. By diagonalizing the matrix $M = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$ write an expression for M^n

Q9. Find a 2×2 example to disprove the statement "All matrices with a repeated eigenvalue can be diagonalized"

Q10. Given $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ is an eigenvector of $M = \begin{pmatrix} 2 & 8 & -2 \\ -2 & 12 & -2 \\ 4 & -4 & 8 \end{pmatrix}$ diagonalize M

Advanced Skills

Q11. Explain why the matrix $M = \begin{pmatrix} a & b-a \\ 0 & b \end{pmatrix}$ isn't diagonalizable and show that $M^n = \begin{pmatrix} a^n & b^n - a^n \\ 0 & b^n \end{pmatrix}$

Q12. Prove for A with eigenvalues λ_1, λ_2 and corresponding eigenvectors v_1, v_2 B with eigenvalues μ_1, μ_2 and corresponding eigenvectors v_1, v_2 that the product matrix $M = AB$ has eigenvalues $\mu_1\lambda_1, \mu_2\lambda_2$ and corresponding eigenvectors v_1, v_2 and hence write D, U when $AB = UDU^{-1}$

Q13. Prove "If two matrices are diagonalizable with distinct eigenvalues and eigenvectors then the two matrices products are equal if the eigenvectors of the two matrices are identical"

Extension

Q14. Can you find any conditions for diagonalization in general

Hint: Think about following the method, when will it always work